

UNRAID

Team 6 — UnRAID NAS

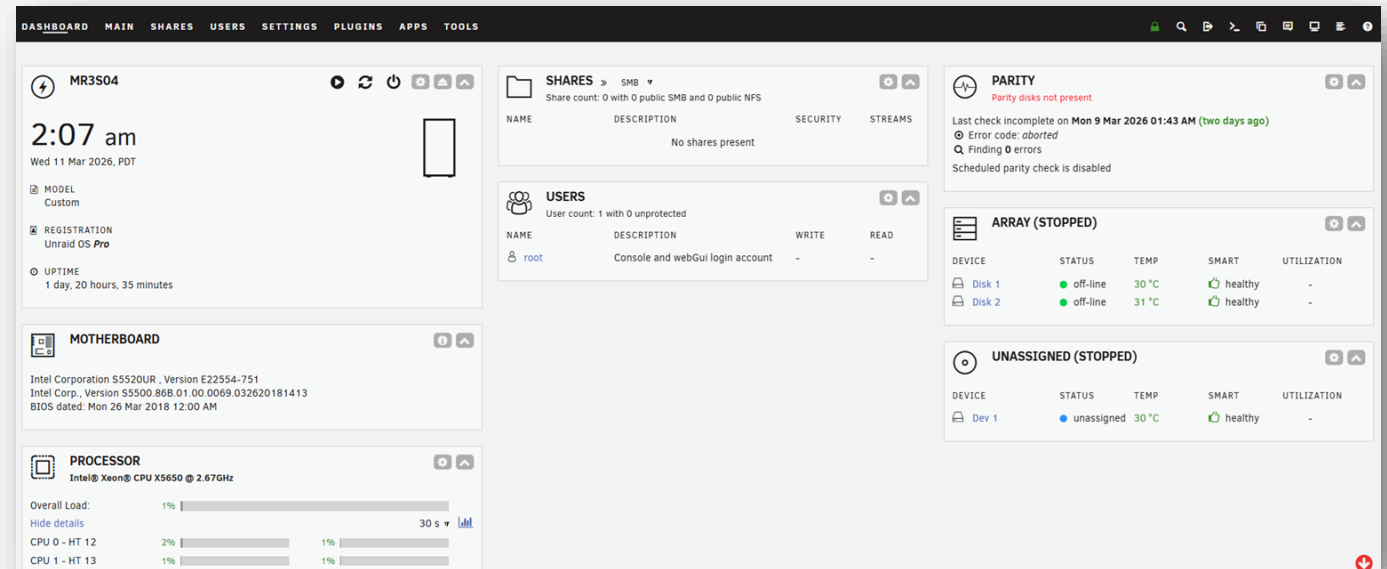
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Server MR3S04 | CLOIF2 | 13.04.2026

Linux-based NAS • Parity Storage • Docker & VMs • Plugin Ecosystem

What is UnRAID?

- Commercial, Linux-based NAS OS by Lime Technology
- Runs entirely from a USB stick — OS never touches data disks
- Licence is hardware-bound to the USB's unique ID
- Web-based management (no CLI needed for daily use)
- Supports Docker containers and VMs natively
- Extensible with community plugins (Community Apps store)
- Own storage model: parity-based, disks stay independent



Only paid NAS OS among the four tested in this class

UnRAID Licensing

Starter	Unleashed	Lifetime
\$49	\$109	\$249
one-time	one-time	one-time
Licence Perpetual	Licence Perpetual	Licence Perpetual
Storage 6 devices	Storage Unlimited	Storage Unlimited
Updates 1 year	Updates 1 year	Updates Lifetime
Support Tickets, forums, docs	Support Tickets, forums, docs	Support Tickets, forums, docs

- All licences are one-time purchases, not subscriptions
- After the first year, update extensions cost \$36/year (optional — not required to keep using UnRAID)
- All tiers include customer support: tickets, forums, documentation, and one-on-one sessions

Source: unraid.net/pricing

Getting Started

Licensed USB

Received from teacher
HW-bound licence key

Clean Config

Cleared previous config
Kept licence key file

Network Setup

Hostname: MR3S04
IP: 10.0.13.4/16

Updated OS

7.1.2 → 7.2.4
Latest stable release

Server name: MR3S04

Description:

Model: Custom

APPLY

DONE

Array must be **Stopped** to change

IPv4 DNS server assignment:	Static	▼
IPv4 DNS server 1:	10.0.0.1	
IPv4 DNS server 2:	1.1.1.1	
IPv4 DNS server 3:		

Interface description: INFO

MAC address:	00:1E:67:09:85:94	
Enable bonding:	Yes	▼
Bonding mode:	active-backup (1)	▼
Bonding members of bond0:	eth0, eth1	▼
Enable bridging:	Yes	▼
Network protocol:	IPv4 only	▼
IPv4 address assignment:	Static	▼
IPv4 address:	10.0.13.4	/ 16 ▼
IPv4 default gateway:	10.0.0.1	1008 ⓘ metric value, a lower value is more preferred
Desired MTU:	1500	☐ Enable jumbo frames ⓘ

Gateway: 10.0.0.1

First Problem: The Web UI Broke

PROBLEM

After config reset, web UI unreachable

nginx returned 502
PHP gave blank page



DIAGNOSIS

Management daemon emhttp wasn't starting after boot



FIX

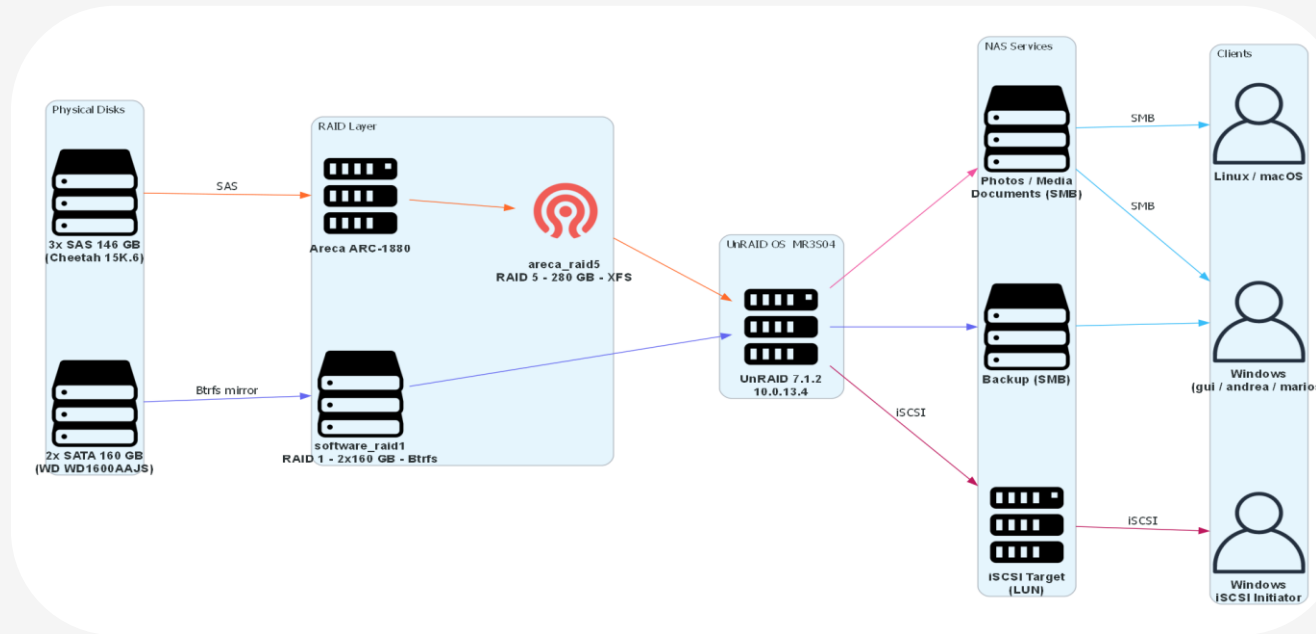
Created /boot/config/go with emhttp startup command

```
/boot/config/go  
/usr/local/sbin/emhttp &
```

The go file is a bash script on the USB that UnRAID executes at every boot — the permanent fix.

Even though UnRAID looks like a simple web UI, underneath it's still Linux — and that's where you fix things.

Storage Architecture



HW RAID 5 — areca RAID5

3x SAS disks, 280 GB total
XFS filesystem
Managed by Areca controller
UnRAID sees it as a single disk

SW RAID 1 — software RAID1

2x SATA 160 GB disks
Btrfs filesystem
Managed by UnRAID
Full mirror — best for critical data

Second Problem: RAID 1 Pool Refused to Start

"Too many wrong and/or missing disks!"

CAUSE

Stale disk assignments from previous team's configuration still in the system



FIX

Tools → New Config
Preserve current assignments: All



RESULT

Reassigned disks,
formatted Btrfs,
RAID 1 pool mounted clean

Two problems, two fixes — both taught us how UnRAID actually works under the hood.

Shares & Access Control

Folder Repartitions

Share	Storage	Purpose
Photos	areca RAID5	Family photos
Media	areca RAID5	Videos, music
Documents	areca RAID5	General files
Backup	software RAID1	Protected backups

User Permissions

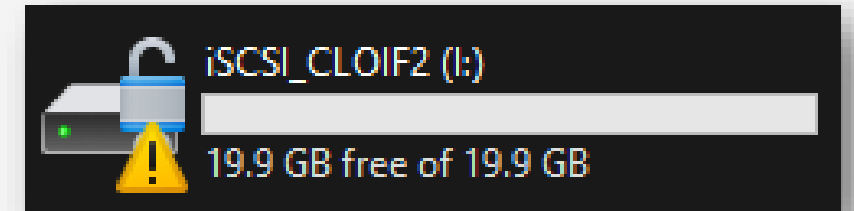
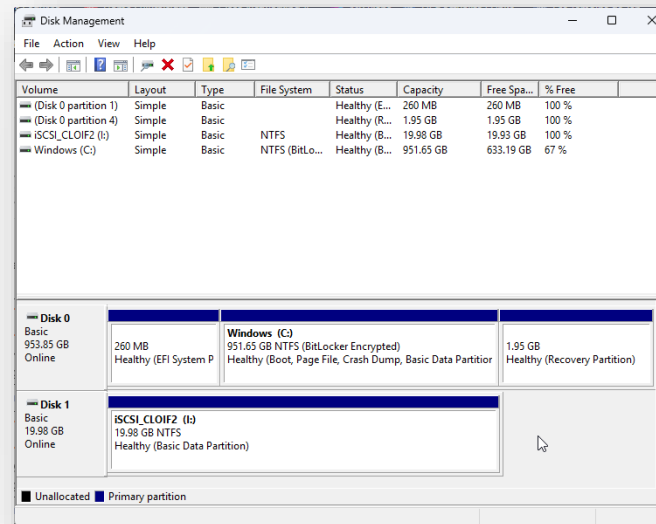
User	Role	Photos/Media/Docs	Backup
gui	Parent	Read/Write	Read/Write
andrea	Parent	Read/Write	Read/Write
marios	Child	Read-only	No Access

All shares: SMB Private — every connection requires username and password (no guest access)

iSCSI: Block Storage via Plugin

iSCSI = the client sees a local disk, not a shared folder — block-level storage over the network

Not built into UnRAID → added via "Unraid iSCSI" plugin (SimonFair)
Created file-based LUN on areca RAID5
Attached to target → authorised Windows initiator



Plugin ecosystem extends UnRAID beyond just file sharing — iSCSI adds enterprise-grade block storage.

Live Demo

- UnRAID dashboard — array status, both storage pools
- Shares and per-user permissions in the UI
- SMB access from Windows (\\MR3S04\Documents)
- Permission denied: marios tries Backup
- iSCSI target + Windows disk (if time allows)

<http://10.0.13.4>

Conclusion

What UnRAID Does Well

- ✓ Intuitive web UI for all management
- ✓ Rich plugin ecosystem (Community Apps)
- ✓ Flexible storage — mix HW + SW RAID
- ✓ Easy share and permission management
- ✓ Docker + VM support out of the box

Where It Falls Short

- ✗ Commercial licence cost (only paid NAS)
- ✗ iSCSI depends on community plugin
- ✗ No native RAID 5 striping
- ✗ Some features need CLI when things break

Sources

unraid.net — Official site & documentation
SimonFair/unraid-iscsi — iSCSI plugin (GitHub)
Areca ARC-1880 — RAID controller documentation
Intel SR2600UR — Server service guide
Course assignment brief — CLOIF2

Thank You

Questions?

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